

Advancing single-cell omics and cell-based therapeutics with quantum computing

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Abstract

The generation of highly accurate models of behaviours of individual cells and cell populations through integration of high-resolution assays with advanced computational tools would transform precision medicine. Recent breakthroughs in single-cell and spatial transcriptomics and multi-omics technologies, coupled with artificial intelligence, are driving rapid progress in model development. Complementing the advances in artificial intelligence, quantum computing is maturing as a novel compute paradigm that may offer potential solutions to overcome the computational bottlenecks inherent to capturing cellular dynamics. In this Roadmap article, we discuss the advancements and challenges in spatiotemporal single-cell analysis, explore the possibility of quantum computing to address the challenges and present a case study on how quantum computing may be integrated into cell-based therapeutics. The specific confluence of quantum and classical computing with high-resolution assays may offer a crucial path towards the generation of transformative models of cellular behaviours and perturbation responses.

Sections

Introduction

Quantum computing in spatial transcriptomics

Temporal quantum computational modelling in single-cell transcriptomics

Integrating single-cell multi-omics and discovering meaningful interactions

Learning from limited samples

A quantum-enabled multimodal cellular model use case

Conclusion and future perspective

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Introduction

Human physiology and diseases are fundamentally a function of the states, behaviours and interactions of billions of individual cells. Therefore, understanding the behaviours of individual cells or cell collectives in a multitude of contexts, such as stimulation through signalling molecules or exposure to therapeutic agents, has wide ranging implications to human wellness, diseases and ageing and is driving technological innovation (Fig. 1). Consequently, technologies such as single-cell sequencing, spatial transcriptomics and spatial proteomics methods have been named 'Method of the Year' by *Nature Methods* in 2013¹, 2021² and 2024³, respectively. Moreover, recent consortia efforts to develop biological atlases, such as The Cancer Genome Atlas⁴, Human Tumour Atlas Network⁵, Human BioMolecular Atlas⁶, Human Cell Atlas^{7,8} and Virtual Cell Atlas⁹, have generated the necessary resources to capture the molecular behaviours of cells and tissues by creating vast repositories of genomic, transcriptomic, proteomic, epigenomic, imaging and so on data. Although each of these modalities is individually informative, the true potential lies in the ability to develop large multimodal models capable of discovering biological properties by integrating the orthogonal information contained across assays (Fig. 1a). These multimodal methodologies are already being deployed for applications such as generating single-cell maps of metastatic breast cancer¹⁰, characterizing tumour evolution within the three-dimensional microenvironmental context¹¹ and building a spatiotemporal atlas of the human maternal–foetal interface¹².

The integration of these high-resolution technologies can capture a holistic perspective of the cellular and metabolic activities of a malfunctioning tissue with single-cell precision, enabling the study of spatiotemporal dynamics of gene expression, molecular interactions, signalling pathways and environmental factors¹³ with unprecedented resolution. Studies demonstrating the clinical usefulness of this perspective include those of cancer¹⁴, cardiovascular disease¹⁵, and diabetes¹⁶. With the advent of spatial transcriptomics methods, it is possible to detect, model and understand cellular microenvironments at exact spatial coordinates. This advancement enables exploring cell–cell interactions (CCIs) while incorporating intracellular and extracellular information to gain insight into the molecular underpinnings of complex diseases¹⁷, whose heterogeneous pathologies cannot be described through single-protein effects.

The breakthroughs made with these assays and consortia efforts were only possible with a corresponding transformative application of artificial intelligence (AI) and machine learning (ML) to biological problems, which enabled new advancements in multimodal models, synthetic biology and protein structure prediction (Fig. 1b). For instance, predicting the three-dimensional protein structure from sequence¹⁸ has been substantially improved through transformer networks¹⁹, with profound effects on synthetic biology and attempts to generate large protein complexes^{20,21}, proteins and, specifically, enzymes^{22,23}. AI/ML models have been used to predict the effects of non-coding genomic variants²⁴, reach human-level performance with automated whole-cell segmentation²⁵ and match the accuracy of clinicians in imaging diagnostics^{26,27}.

Computational methods used in spatial transcriptomics perform multiple tasks, including integration of single-cell RNA sequencing (scRNA-seq) data, cell-type annotation, cell segmentation, spatial clustering, dimensionality reduction, inference of cell trajectories and interactions and data visualization²⁸. These tasks are often performed using methods that can be broadly grouped together on the basis of their underlying similarities, including graph-based deep learning

and probability-based Bayesian methods, kernel methods, optimal transport, tensor decomposition, ensemble learning, non-negative matrix factorization, clustering using kernel methods and topological data analysis (TDA) and so on¹⁷. Although applications of these methods to single-cell analyses have been impressive, important limitations remain that hinder their application in the clinic²⁹. Some of these limitations are related to native properties of single-cell data, e.g. innate complexity and scale²⁸, whereas others are associated with shortcomings of AI algorithms, for example, inability to learn in sample-limited contexts, model overfitting, learning saturation³⁰, difficulty in capturing global contextual signals, poor generalization to out-of-distribution (OOD) data and quadratic time and space complexity, among others. These challenges can be exacerbated in combined multimodal settings, where AI models often fail to capture fine-grained spatial resolution³¹, to generalize across batches or experimental conditions³² or to resolve subtle temporal or spatial trajectories³³.

Though AI has proven crucial for analysing (spatial) single-cell data, the above noted limitations may prevent the full realization of these spatial single-cell technologies. For certain classes of computational problems, the solution may lie with quantum computing, which is an emerging paradigm of high-performance computation that departs from the conventional binary logic underpinning classical computing (Fig. 1b). Instead, quantum computing leverages principles of quantum mechanics^{34,35}, such as superposition and entanglement, to perform calculations. These principles are particularly attractive for those problems whose complexity or memory requirements grow exponentially.

Several different quantum computing architectures are being developed on distinct types of qubits, including neutral atom qubits³⁶, spin qubits³⁷, topological qubits³⁸, trapped-ion qubits³⁹ and superconducting qubits^{40–42}. The scale of these quantum devices ranges from tens to 1,000 qubits, and although they mainly remain susceptible to noise⁴³, the development of error correction techniques used to protect quantum information is advancing^{44,45}. These noisy devices are termed pre-fault tolerant quantum devices (PFTQD), as opposed to fault-tolerant quantum devices, in which error is corrected to yield logical qubits⁴⁵. Progress has been made beyond qubit counts in quantum hardware⁴³, such as dynamic circuits with mid-circuit measurements⁴⁶, improved fidelities of 1-qubit and 2-qubit gates⁴⁷ and increased execution speed of quantum circuits⁴⁷. Moreover, advances in error mitigation and error suppression techniques, when coupled with circuit cutting and knitting techniques^{48,49}, have enabled researchers to scale up the size of their quantum experiments.

These quantum computation advancements (Fig. 1b) demonstrate progress towards a stage where quantum computers can solve certain meaningful problems faster, cheaper or more accurately than classical computers alone for selected applications^{40,50,51}. Similarly, quantum algorithms such as quantum optimization and quantum ML (QML) aim to analyse datasets more efficiently or with better generalization than classical algorithms^{52–54}. Depending on the specific use case, quantum algorithms may enable certain benefits in metrics such as accuracy, energy efficiency, input data requirements and speed⁵⁵.

Driven by the rapid advancement in both quantum computing and single-cell technologies, in this Roadmap, we consider the potential of quantum computing to accelerate discovery in single-cell and spatiotemporal single-cell analyses, particularly given certain constraints faced by classical computing^{56–60}. In the following five sections, we discuss the advantages of and challenges faced by

Roadmap

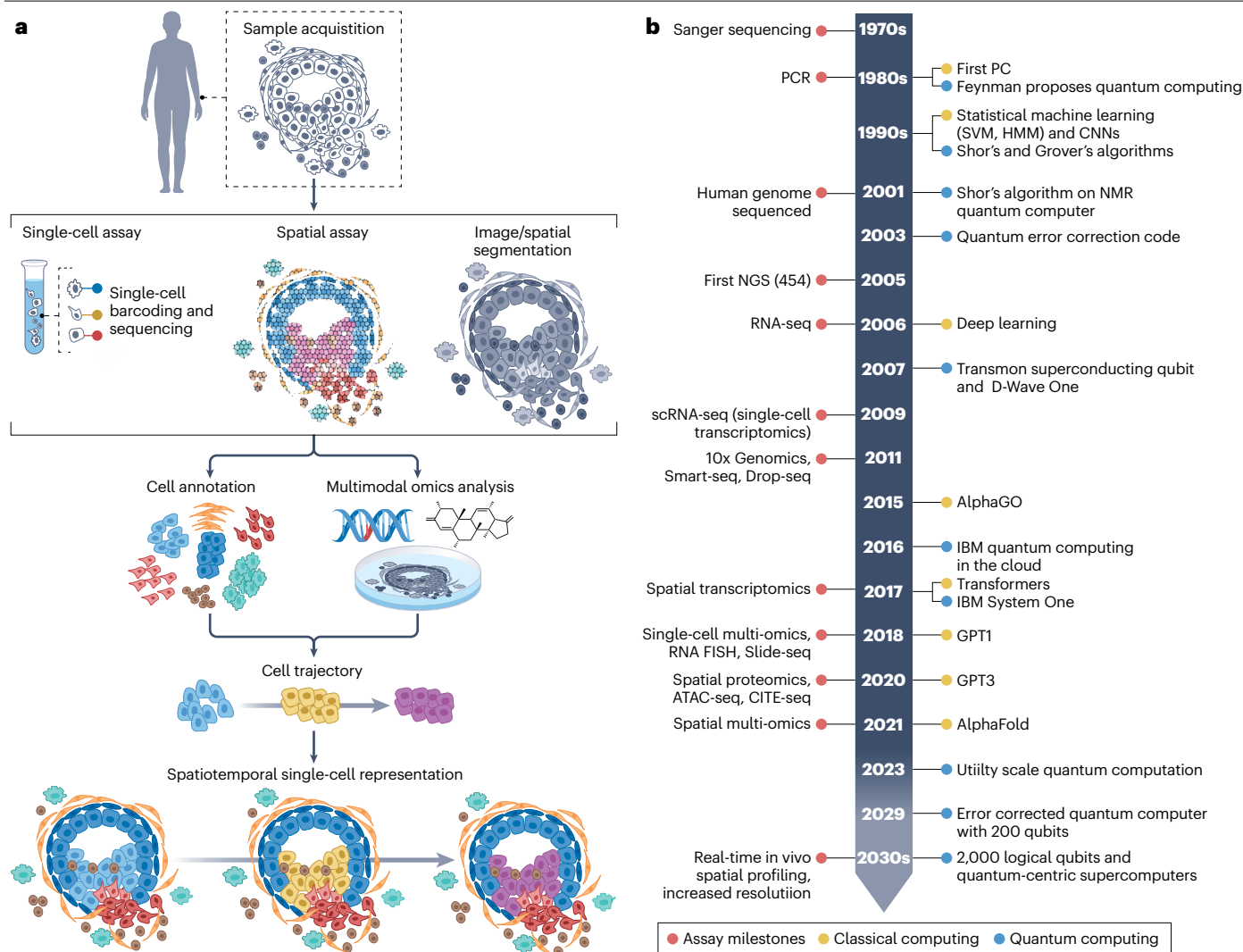


Fig. 1 | Overview of spatial and single-cell assays and relevant quantum computing techniques. **a**, Single-cell and spatial data are studied through a variety of techniques, including profiling of biomolecules in single-cell assays; spatial assays that capture regions where biomolecules are present (for example, single-molecule fluorescence in situ hybridization); image and spatial artificial intelligence segmentation to detect cells, cellular structures and tissues from imaging and spatial data; cell annotation; multimodal omics data integration; and cell trajectory analysis, with the goal of creating a spatiotemporal representation with single-cell resolution. **b**, A timeline of relevant milestones

in the development of biological-assay development, classical computational achievements and quantum computational achievements. ATAC-seq, assay for transposase-accessible chromatin using sequencing; CITE-seq, cellular indexing of transcriptomes and epitopes by sequencing; CNN, convolutional neural network; GPT1, generative pretrained transformer 1; HMM, hidden Markov model; NGS, next-generation sequencing; NMR, nuclear magnetic resonance; PC, personal computer; RNA FISH, RNA fluorescence in situ hybridization; scRNA-seq, single-cell RNA sequencing; SVM, support vector machines.

state-of-the-art algorithms in multiple domains and review quantum computing analogues for those techniques, in addition to discussing how they may be able to solve some of the identified challenges (Fig. 2 and Table 1). We address topics of quantum computing in spatial transcriptomics, temporal quantum computational modelling in single-cell transcriptomics, integration of single-cell multi-omics and learning from limited numbers of samples. We finally provide a framework for integrating quantum algorithms with state-of-the-art ML algorithms for a quantum-enabled multimodal cellular model use case (Fig. 3).

Quantum computing in spatial transcriptomics

We will use spatial transcriptomics as a specific data modality to highlight computational tasks that would be applicable to similar modalities such as spatial proteomics. Spatial transcriptomics refers to methodologies for quantifying mRNA expression of large numbers of genes in addition to preserving the positional context of cells in a tissue. This is achieved through either imaging-based methods, such as single-molecule fluorescence in situ hybridization (smFISH), which visualizes gene expression in the cellular space, or through sequencing-based methods that capture transcriptomes from defined tissue regions⁶¹.

Methods for spatial data processing

Analysing spatial transcriptomics data entails both upstream and downstream data analysis. Upstream, raw data from smFISH-based techniques are processed to identify transcript spots, match spots to genes and assign spots to cells⁶¹. Achieving cellular resolution with high-density measurements requires accurate cell segmentation and manual fine-tuning with corrections⁶². Various classical computation methods have been proposed for cell segmentation, including conventional deep learning methods and techniques that integrate scRNA-seq data. For example, CelloType⁶³ utilizes a transformer-based deep learning technique for multiscale cell segmentation and annotation of spatial omics data. Methods such as Baysor⁶⁴ spatially cluster observed molecules to cells from smFISH data by modelling each cell using Markov random fields. The method spatial transcriptomics cell segmentation⁶⁵ with its transformer-based model infers each spot as either a part of a cell or the surrounding extracellular matrix through an attention mechanism and works around the rigorous hyperparameter tuning of conventional image segmentation methods by incorporating scRNA-seq data. These methods are usually constrained by the quality and volume of training images and scRNA-seq data, the accurate integration of high-dimensional transcriptomics and imaging data, and poor generalization in OOD datasets and cell types⁶⁶. Typically, smFISH techniques produce low spatial resolution data. To address the absence of transcriptome-wide expression values for unprofiled genes, these datasets often employ reconstruction methods that frequently leverage scRNA-seq data along with a select group ‘landmark’ genes⁶⁷. Most of these methods either use similarity measures between each cell

in scRNA-seq and each location, favouring genes that are specific to a subset of cells in a spatial reference⁶⁷ or formulating the reconstruction as an optimal transport problem for probabilistic embedding of similar cells⁶⁸.

Many quantum algorithms relevant to upstream smFISH preprocessing tasks, such as for cell segmentation (Fig. 2 and Table 1) are being actively studied for their potential utility on quantum devices. For example, quantum convolutional neural networks (QCNN)^{69,70} may be a promising alternative to transformer-based methods such as CelloType owing to potential greater generalizability for accurately segmenting unknown cell types. Alternatively, such transformer-based methods may be substituted with quantum doubly stochastic matrix (DSM)-based transformers that replace the conventional softmax function in the attention module, which yields a right-stochastic matrix, with a variational quantum circuit to compute a DSM that has shown certain advantages⁷¹. Similarly, quantum algorithms have been proposed with theoretical polynomial speedup over their classical counterpart in learning Markov random fields⁷². Recently, quantum transformer models have been proposed for performing natural language processing tasks requiring fewer parameters compared to the classical benchmarks^{73,74}, which can potentially be leveraged in cell segmentation tasks being performed by methods such as spatial transcriptomics cell segmentation. For spatial reconstruction of smFISH data, identifying the similarity between cells and spatial locations can be challenging as functional regions of tissues may not be composed entirely of cells with identical expression patterns⁶⁷. Representing cell similarity as a graph of cells and employing discrete-time quantum

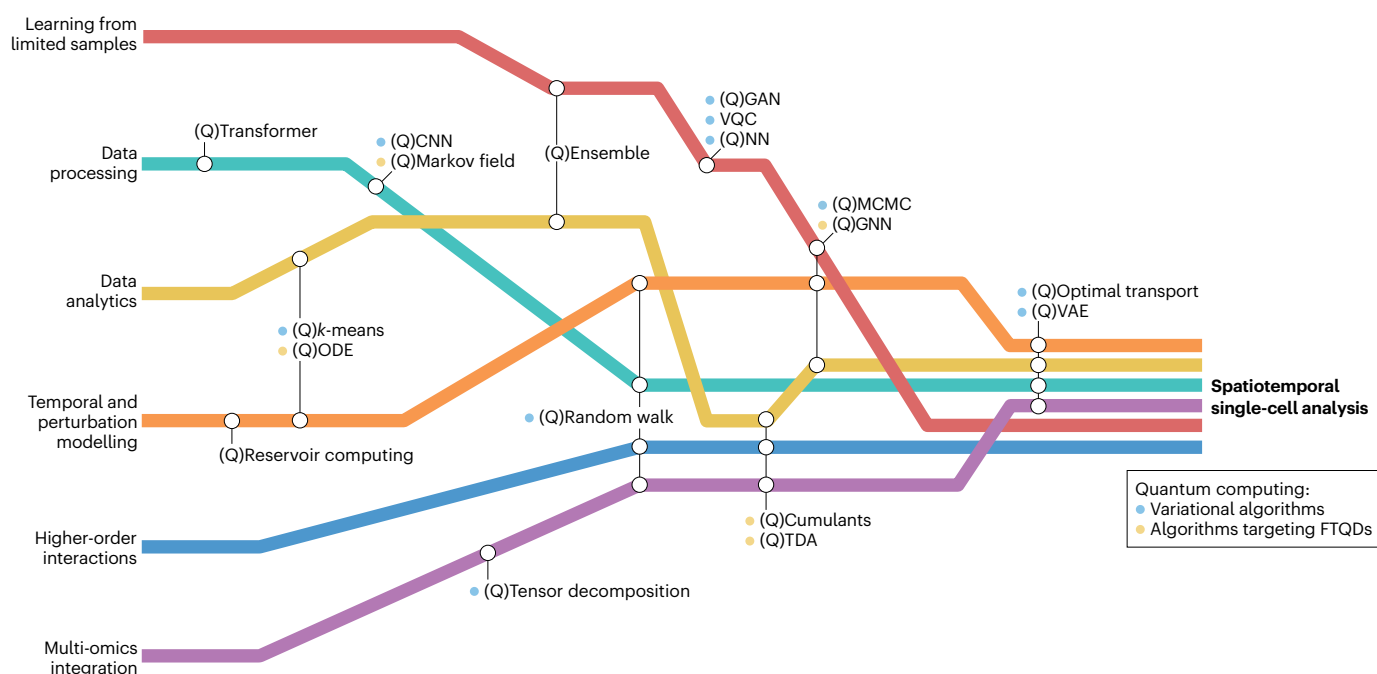


Fig. 2 | Map of computational tasks performed by classical and quantum computation algorithms for spatiotemporal single-cell analysis. These six broad computational tasks (coloured lines) are shown intersecting with various relevant applicable algorithms denoted by circles, all leading towards the goal of spatiotemporal single-cell analysis. Certain classical methods have a quantum analogue – denoted with a preceding (Q). These algorithms and their considerations are described in further detail in Table 1. FTQDs, fault-tolerant

quantum devices; QCNN, quantum convolutional neural network; QCumulant, quantum cumulant calculation; QGAN, quantum generative adversarial network; QGNN, quantum graph neural networks; QMCMC, quantum Monte Carlo Markov chain; QNN, quantum neural network; QODE, quantum ordinary differential equations; QTDA, quantum topological data analysis; QVAE, quantum variational autoencoders; VQC, variational quantum circuit.

Roadmap

random walks⁷⁵ to find trajectories of similar cells can be an alternative for capturing fine-grained mappings of cell locations with scRNA-seq data. For optimal transport mapping of the cell spatial reconstruction problem, the quantum conditional optimal transport method can be used to effectively map the problem into a quantum circuit and solve the assignment problem⁷⁶.

Methods for spatial data analytics

A host of downstream classical computing methods have been developed to analyse spatial data collected by upstream techniques. Many popular scRNA-seq packages provide such a capability, including Seurat⁷⁷, Squidpy⁷⁸, Giotto⁷⁹ and so on. In downstream processing of spatial transcriptomics data, some of the major areas of analysis

Table 1 | Classical computing algorithms, their challenges and their quantum computing analogues in performing various tasks in single-cell analyses

Classical computing method	Challenges	Quantum computing equivalent	Biological applications
Convolutional neural network	Requires large data and computational resources to train; weak generalization and unstable transfer learning	(Quantum) convolutional neural network ^{69,70}	Cell classification ²⁰⁹ and segmentation ⁶² ; CpG prediction ²¹⁰ ; T cell receptors classification ²¹¹
Markov random field	Learning latent structures in graphs; curse of dimensionality	(Quantum) Markov random field ^a (ref. 72)	Cell segmentation, clustering and cell-type annotation ⁶⁴
Transformers	Requires large data and computational resources to train; poor OOD generalization; lack of accuracy in noisy data	QTransformer ^{71,73,74}	Cell segmentation ⁶³ ; single-cell DNA sequencing denoising and imputation ²¹² ; CpG prediction ²¹³ ; perturbation modelling ^{134–137}
Random walk	Sensitive to restart probability; difficult to select parameters; computationally expensive; limitations in detecting communities	Discrete-time and continuous-time random walks ^{75,163,164}	Cell clustering and imputation ²¹⁴ ; cell trajectory inference ¹⁵² ; CCI ²¹⁵
Optimal transport	Sensitive to outliers; computationally expensive in large datasets; sensitive to optimization parameters; requires large samples to accurately estimate transport map	Quantum optimal transport ²¹⁶ ; quantum conditional optimal transport ⁷⁶	Spatial reconstruction ⁶⁸ ; cell differentiation ¹³³ ; lineage tracing ²¹⁷ ; trajectory inference ²¹⁸ ; cell–cell communication ²¹⁹ ; perturbation modelling ^{132,133} ; multi-omics integration ²²⁰
Graph neural network	Addressing class imbalance; sensitive to noise in graphs; OOD generalization; transfer learning issues	Quantum graph neural networks ²²¹	Cell-type clustering ⁸⁹ ; deconvolution ⁹⁰ ; CCI ⁹¹ ; spatial transcriptomics ⁸⁸
Markov chain Monte Carlo	Requires extensive hyperparameter tuning; sensitive to convergence criteria; high autocorrelation; takes long time to converge	Quantum Markov chain Monte Carlo ^a (ref. 95)	Spatial transcriptomics ⁶⁷ ; perturbation modelling ¹¹⁶
Ensemble learning	Computationally expensive; weighting each member in ensemble; determining optimal number of base learners; hyperparameter tuning ²²¹	Quantum ensemble learning ^{96,98}	CCI ⁹⁰ ; classifying single-cell data; differential gene expression in single-cell RNA sequencing ²²² ; prediction tasks ²²³
Topological data analysis	Does not scale for larger datasets and higher order; interpretability; parameter selection	Quantum topological data analysis ^a (refs. 99–101)	Spatial transcriptomics ⁸⁷ ; multimodal integration ¹⁷²
k-means	Parameter selection; sensitive to outliers; scaling to large datasets; cluster size sensitivity	Quantum k-means ²²⁴	Cell trajectory inference ¹⁰⁹ ; cell clustering ²²⁴
Gene regulatory network	Unable to infer simultaneous, inter-regulatory connections between all genes in concert	Quantum gene regulatory network ²²⁵	Perturbation modelling ³³ ; predicting transcription factors ²²⁶
Variational autoencoder	Learns uninformative latent representations; sensitive to noisy data; selecting hyperparameters; lack of OOD generalization	Quantum variational autoencoder ^{148,149}	Trajectory inference ¹²² ; perturbation modelling ¹³⁶ ; multi-omics integration ¹⁵⁷
Continuous-state hidden Markov model	Estimation of posterior state probabilities; scales exponentially with large datasets; assumes conditional independence in input states ²²⁷ ; sensitive to convergence criteria	Quantum circuits for graphical model or quantum probabilistic graphical models ^a (ref. 139)	Trajectory inference ¹¹⁶
ODE	Stiffness; parameter estimation; scaling issues with larger datasets	Quantum algorithms for solving ODE ^a (ref. 227)	Cell differentiation ^{112,113,115} ; trajectory inference ²²⁸
Neural ODE	Computationally expensive; stiffness; lack of OOD generalization ²²⁹	Quantum neural ODE ¹⁵¹	Cell differentiation ¹¹⁵ ; trajectory inference ²³⁰ ; temporal modelling ^{150,229}
Reservoir computing	Difficult to estimate nonlinear functions; computationally expensive	Quantum reservoir computing ^{140,142,143}	Temporal cell segmentation ¹¹⁸
Generative adversarial network	Poor OOD generalization; non-convergence and instability; vanishing gradients ²³¹	Quantum generative adversarial networks ^{144–146,178,180}	Perturbation modelling ^{135,137} ; trajectory inference ²³² ; single-cell RNA sequencing imputation ²³³
Tensor decomposition	Intractable for large datasets and higher orders; determining optimal rank; nonlinear feature extraction	Quantum tensor decomposition ^{173–175}	Multi-omics integration ¹⁵⁶ ; spatial transcriptomics ²³¹

^aImplementation of fault-tolerant quantum devices required. CCI, cell–cell interactions; ODE, ordinary differential equation; OOD, out of distribution.

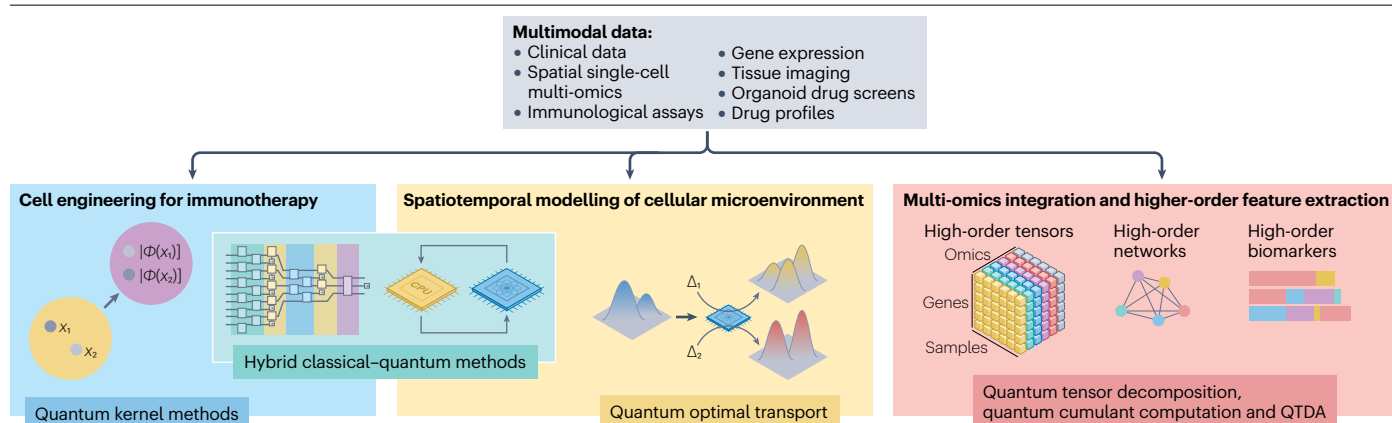


Fig. 3 | Overview of a quantum-enabled cell-based therapeutics.

Spatiotemporal single-cell, cell population, imaging, drug profile and clinical data are analysed with three different goals enabled by quantum computing technologies to capture varying aspects of cellular behaviours. These technologies include (from left to right): hybrid classical-quantum machine learning with quantum convolutional neural networks or quantum kernel methods, integrated with classical central processing units (CPU), to aid cell engineering for immunotherapy, hybrid classical-quantum machine learning and quantum optimal transport to model tumour microenvironments from single-cell spatial data with perturbation modelling to predict cellular response, and multi-omics integration and higher-order feature extraction to capture

cryptic interactions and biomarkers of single-cell perturbation responses using quantum tensor decomposition, quantum cumulant computation, and quantum topological data analysis (QTDA). These higher-order biomarkers are represented by different numbers and combinations of features from various modalities. Using insights from each research area and by combining the efforts between them, a cell-centric therapeutics design strategy can be achieved for treatment options that optimize the cellular context to improve therapeutic response. x_1 and x_2 represent the original data vector for samples 1 and 2, respectively. $\Phi(x_1)$ and $\Phi(x_2)$ represented the quantum embedding of the x_1 and x_2 data vectors, respectively. Δ_1 and Δ_2 represent different trajectories of the cellular state space.

involve capturing alterations in gene expression, detecting mutations, examining associations between gene expression and cell-type localization, and identifying spatial regions. These tasks are often done by modelling and learning on spatial cell graphs in the form of graph neural networks (GNNs), explicitly learning on cells instead of encoding pixels and integrating single-cell information with disease tissue (for example, tumour) morphology and topology, as well as interactions among cells and tissues⁸⁰. GNNs are constructed on these cell graphs with an interpretable input space. When paired with model explainability techniques, they can provide valuable insights into which cells or cell neighbourhoods drive the biological process in tissues⁸¹. GNNs are ubiquitously used in spatial transcriptomics tasks such as cell-type clustering⁸² and deconvolution⁸³ in the upstream and predicting spatially resolved gene expression or modelling cell microenvironment with cell-tissue interactions and CCIs⁸⁴ in the downstream. However, the complexity of tumour graphs and the entangled cell neighbourhoods can lead to suboptimal embedding spaces of GNNs, hampering the learning of clinically meaningful patterns. Moreover, searching for relatively small query subgraphs over large, complex graphs is nondeterministic polynomial-hard⁸⁵. Other methods to find spatially resolved genes commonly use Gaussian process regression⁸⁶, TDA⁸⁷, spatial autocorrelation⁸⁸ and more. Most of these methods lack statistical power, require extensive hyperparameter fine-tuning²⁸ or are time consuming, particularly those employing Bayesian methods such as Markov chain Monte Carlo (MCMC). Similar to resolving spatial regions, characterizing CCIs are computationally expensive tasks often achieved by identifying ligand-receptor pairs⁸⁹. As gene expression is affected by CCIs, several packages have been developed to account for the association of ligand-receptor pairs with gene expression by constructing a covariance matrix of neighbourhood graphs⁹⁰ or using ensemble learning of expression of each gene of interest⁹¹. However,

such approaches necessitate computationally heavy operations, such as inverting large matrices or computing pairwise distances for large numbers of cells.

These limitations present opportunities for quantum computing, for example, in classification of tumours or optimizing subgraph matching using GNNs (Table 1). Quantum-classical computing hybrid methods can leverage the state-space available to quantum methods as well as benefit from state-of-the-art classical GNN models for spatial transcriptomics tasks, to improve predictive performance for complex diseases through spatial reconstruction or modelling tumour microenvironment. Algorithms such as variational quantum classifiers (VQCs), quantum neural networks (QNNs) and others have shown to be more generalizable when learning from a relatively smaller training dataset⁵². Thus, these models can be used to learn from a low-dimensional embedding in small sample size scenarios, as is often the case in spatial transcriptomics given limited availability of whole slide images and fewer genes as reference⁹². A hybrid quantum-classical model, which used an end-to-end training paradigm from GNN embeddings of whole slide images and VQC, achieved comparable performance to baseline methods using GNNs for breast cancer subtyping⁹³. As quantum computing hardware is still evolving, efficient strategies for subsampling, such as obtaining coresets⁹⁴ or kernelized subsampling methods, are often used to fit the training data into quantum hardware for efficient processing and discerning disease subtypes⁹². To address the issue of the scalability of Bayesian methods for identifying spatially resolved genes, quantum MCMC, which has been empirically shown to require fewer iterations to converge and is more robust to noise⁹⁵, can be applied. Similarly, quantum ensemble methods can make comparable predictions with a magnitude fewer number of learners or samples^{96,97} and in some cases achieves a quadratic improvement over classical ensemble learning methods such as AdaBoost⁹⁸. These methods can

improve the performance of the ensemble learning method to extract relationships between spatial omics data⁹¹. Quantum TDA (QTDA) has demonstrated a super-polynomial speedup in fault-tolerant quantum devices for some particular structured graphs with large spectral gaps in the Laplacian⁹⁹ or in some cases provable exponential speedup¹⁰⁰. Hence, QTDA can be used to identify spatially resolved genes by fast computations of the Betti numbers^{87,101}.

Temporal quantum computational modelling in single-cell transcriptomics

Biological processes are often better understood through a temporal lens, but scRNA-seq datasets are typically collected at different time points, then integrated for analysis. In some cases, these time points are hourly, as in immune response and drug treatment studies, whereas in other cases they can span days, months or years, as in cell differentiation and clinical trials¹⁰². Statistical, graph-based and ML techniques are then applied to examine cell frequencies within samples, delineate gene regulatory networks (GRNs) in individual cells and capture relationships among cells. Although capturing dynamics of cells is naturally achieved using time series, it can also be performed by analysing a snapshot from a single sample using methods such as trajectory inference or RNA velocity¹⁰³.

Trajectory inference is often done by first embedding the scRNA-seq data to a lower dimension with principal component analysis, diffusion maps, autoencoders and so on³³. Then, a trajectory tree is inferred, learning on a lower-dimensional manifold, which captures the underlying biological variation. A root cell is then selected on the basis of a prior or data-driven heuristics, and pseudotime values are assigned to all other cells according to their positions along the inferred trajectory^{104,105}. Cell expression similarity and entropy¹⁰⁶, which are free of root cell determination, are also used in this step. Popular trajectory inference methods infer trajectory graphs and cluster cell states by either graph partitioning methods¹⁰⁷, minimum spanning trees¹⁰⁸, or clustering using *k*-means followed by pruning small branches and leveraging spatial information¹⁰⁹. Methods based on optimal transport theory^{110,111} that focus on a distribution of cell placements along the trajectory rather than on specific placement, have also been developed. The continuous dynamics of cell differentiation can be reconstructed from snapshot measurements by leveraging connections of optimal transport with ordinary differential equations (ODE)^{112,113}, but they are limited by either ineffective dimensionality reduction techniques or model limitations¹¹⁴. In addition, a combination of variational autoencoders (VAE) with neural ODE have been proposed to overcome these limitations¹¹⁵. Continuous-state hidden Markov models (HMMs) are also used to assign cells to any position in its trajectory by initiating the assignment using clustering and thereafter assigning them with greater assignment probability in each step of the iteration, until a criterion is met¹¹⁶. Reservoir computing is another means for modelling information dynamics in cells¹¹⁷. Reservoir computing models are based on recurrent neural networks to project input data into higher dimensional spaces based on the dynamics of a 'reservoir', which is a fixed, nonlinear system. Such methods are well suited to temporal or sequencing processing, showing improved performance for temporal segmentation of T cells from high-resolution live cell imaging¹¹⁸.

Perturbation modelling

Modelling dynamic behaviours of cells is not limited to lineage tracing and interactions: there is also a need to study how external influences, such as genetic knockouts, drug interventions, temperature

and so on, perturb cells, to better understand pharmacological and translational aspects of cellular processes¹¹⁴. Modelling perturbation responses at single-cell resolution is imperative for designing novel therapeutics^{119,120}. Perturbation modelling can be categorized into five main objectives: predictive models of changes in cellular behaviour; discovering phenotypic characteristics of cells under perturbation; predictive models of perturbation by drug compounds, which facilitate drug repurposing; identifying synergistic or antagonistic interactions; and predicting chemical structural components from perturbation data¹¹⁴. These objectives can then be addressed by a variety of classical and quantum algorithms (Fig. 2 and Table 1). Technologies built on CRISPR-mediated pooled knockout screens such as Perturb-seq¹²¹ and its variants such as CRISP-seq¹²¹ provide the data to understand a gene's function with respect to its expression when knocking out or knocking down a gene. These perturbation-based technologies have been developed to capture perturbation responses on single-nucleotide polymorphisms with Mix-seq¹²², perturbations in multi-omics modalities with Perturb-CITE-seq¹²³ and perturbations in epigenetic and transcriptome mosaics with Mosaic-seq¹²⁴.

Classical computational tools that analyse single-cell CRISPR screening data range from statistical inference models to complex generative and AI foundation models¹²⁵. The statistical graph-based models are often probabilistic and regression-based and provide dimensionality reduction¹²⁶, clustering¹²⁷ and robust statistical analysis after removing confounding effects. GRNs are leveraged to make predictions such as identifying transcription factors controlling genes by integrating scRNA-seq and scATAC-seq (assay for transposase-accessible chromatin using sequencing) data¹²⁸. Generative models are used ubiquitously in predicting perturbation responses given the availability of large-scale atlases of cells. Methods such as generative adversarial networks (GAN) have been proposed for simulating single-cell differentiation through latent spaces¹²⁹. However, due to constraints of GANs when training on high-dimensional structured data, and effective mapping of gene expressions to latent spaces, VAEs¹³⁰ have been used more frequently in generating cells with desired properties. For instance, scGen combines VAEs and latent representations of a low-dimensional manifold to capture underlying cellular mechanisms while predicting perturbation responses within and between cell types¹¹⁹.

A recent trend is using the mathematical framework of optimal transport or cutting-edge transformer models to capture perturbation effects in scRNA-seq data¹³¹. Optimal transport methods such as CellOT¹³² and CondOT¹³³ provide a model for molecular profiling of cells by generating an optimal transport map for each perturbation to accurately predict cell-to-cell variability in drug responses, drug's mode-of-action, OOD extrapolations across species and extracting subpopulation-specific drug effects. Foundation models with pretrained transformers have also been developed for understanding the dynamics of cellular responses to *in silico* perturbation to predict phenotypes¹³⁴, cell-type-specific transcription factors¹³⁵, drug response¹³⁶, cell similarity¹³⁷ and potential therapeutic targets. Although the advent of the transformer architecture and large language models provide exciting opportunities, they have many limitations ranging from the quality of data they are trained on, effectiveness of the prediction, ability to capture underlying dynamics in GRNs, robustness of multitask learning and model interpretability^{119,120}.

There are several opportunities for quantum computing to enhance perturbation modelling. HMMs can be represented as probabilistic graphical models – a chain of dependent states with inference based on given outcomes. This makes HMMs suitable for time-series

analysis, and continuous-state HMMs have been proposed to derive cell state trajectory¹¹⁶. However, their applications face difficulties when there are structures with high-order interactions, which is often the case for genes affecting cellular trajectories across space¹³⁸, leading to computational intractability. Recently, quantum circuits for graphical models have been proposed to obtain an unbiased estimate of the underlying probability distribution of a graphical model¹³⁹. Quantum circuits for graphical models scale in the worst-case exponentially in the number of model parameters, which can be a substantial improvement over classical probabilistic graphical models, such as MCMC or HMM. Quantum reservoir computing and its many variants have also been proposed to estimate nonlinear functions of an input quantum state to understand the dynamic process influencing it¹⁴⁰. The training is conceptually simple and computationally inexpensive¹⁴¹, making quantum reservoir computing attractive for applications in modelling cellular dynamics. These reservoir computing systems can be made more computationally efficient through physical implementations¹⁴², for example, photonic reservoir computing and quantum mechanical implementations¹⁴³. Quantum GANs^{144–146} have been proposed as an alternative to GANs as they are able to learn underlying probability distributions with fewer parameters, making them useful in perturbation modelling. Similarly, quantum VAEs with quantum circuits as autoencoders and classical data compression^{147,148} or with latent generative processes implemented in a quantum Boltzmann machine have been implemented and applied on classical benchmarking datasets such as MNIST¹⁴⁹. Quantum VAE can then be leveraged to potentially improve latent representations of cells in generative models for perturbation modelling.

Optimal transport has fuelled many applications in spatiotemporal modelling of single cells, including inferring cellular trajectory, dynamic modelling of cells, perturbation modelling and more, by learning a global transport map between cell states. Recently, a quantum computing formulation of the amortized optimization of interpretable contextual transportation maps was proposed for predicting changes in the composition of a cell population due to drug perturbations⁷⁶. In this case, a link between unitary operators and DSMS is leveraged to directly compute optimal transport maps through a quantum circuit. In another application, neural ODEs are proposed in conjunction with optimal transport methods to reconstruct dynamics of cell differentiation¹⁵⁰. Quantum dynamics latent neural ODE was designed to learn and generate quantum trajectories of open and closed quantum systems¹⁵¹. Cell state trajectory can be represented as a graph¹⁵² and encoded into quantum states using quantum feature maps¹⁵³, and quantum dynamics latent neural ODE can then be used to infer trajectories with high accuracy. These quantum equivalents of classical methods have the potential to solve the limitations of classical methods such as lack of generalizability, interpretability and tractability, among others.

Integrating single-cell multi-omics and discovering meaningful interactions

Single-cell transcriptomics is often integrated with other omics data such as genomics, phenomics, epigenomics, proteomics, metabolomics and so on to gain a holistic understanding of cell and tissue function¹⁵⁴. Integration of different omics modalities is imperative to achieve this goal; however, owing to incompleteness of information across omics, technological throughput constraints, measurement variances between devices or simply lack of better technology to measure molecular entities, there is a need for novel methods to better explore

cryptic relationships, probably residing in topological connections among important higher-order interactions between the dimensions¹⁵⁵. Such methods, which leverage manifold alignment, optimal transport and tensor decomposition have been proposed to align and integrate multimodal, single-cell multi-omics^{155,156} (Fig. 2 and Table 1). Other methods using canonical correlation analysis or non-negative matrix factorization find a shared biological space in a lower-dimensional embedding of different omics modalities from different cells to integrate datasets⁷⁷. Graph-based embedding techniques have also been employed for multi-omics integration using graph VAEs to learn embeddings and then reconstruct and link omics data by aligning cellular embeddings¹⁵⁷. Integrating multi-omics data to learn GRNs and infer results of perturbations while predicting transcription factors provides new opportunities to understand the function of cells within tissues¹⁵⁸.

Network and graph representations have been extensively applied to single-cell, spatial transcriptomics and molecular interaction data whether as internal representations for deep learning models or as cellular and molecular networks^{159,160}. These data have given rise to so-called quantum network medicine¹⁶¹ principles, which are typically used to investigate the molecular complexity of a particular disease or the molecular relationships among apparently distinct phenotypes from network topologies. However, such networks are vastly incomplete, with large proportions of the true interactions being still unknown¹⁶². To efficiently search for new drugs, drug combinations, host–pathogen interactions and so on within and between cells, computational methods are required that can access the immense, largely unexplored molecular space. Quantum computing may be a key ingredient in enabling the full potential of network medicine (Table 1) through quantum network medicine¹⁶³. This approach has already proven successful in performing link prediction based on continuous-time quantum walks¹⁶³ and has been successfully adapted to identify disease modules and new disease pathways¹⁶⁴.

TDA and cumulants extend the analysis of data topologies to capture hidden structures from the shape of data and from d th order interactions representing complex associations, leading to logical relationships in phenotypes, for example, interactions between collectives of cells, features or alterations that lead to therapeutic response¹⁶⁵. TDA is used in computer vision and atomistic models, and cumulants of order $d > 3$ have important roles, including in financial data analyses and hyper-spectral image analyses. Cumulants are also useful for integrating and analysing multi-omics data^{165,166}, whether single-cell or patient omics data, where they can provide a unique measurement of liability or risk of a given disease or trait, for each cell or individual¹⁶⁷. The distance between meta-features from cumulants can be calculated and TDA applied to generate the topological features that characterize the clusters of features¹⁶⁸. This methodology aids in uncovering intricate patterns and relationships that may be elusive through conventional statistical methods. Given strong correlations in healthcare data, high-order cumulants help filter out spurious low-order effects that vanish when variables are independent¹⁶⁹, leading to successful applications in constructing aggregated phenotypes that represent distinct pathways in complex diseases^{165,170}.

QTDA and quantum cumulant calculations can potentially detect features and higher-order interactions efficiently, especially for regression and classification tasks (Table 1). Traditional challenges in TDA and cumulant computation come from the exponential growth of possible feature combinations in high-dimensional data. The first QTDA algorithm specifically targeted situations with high-dimensional features and provided an exponential speedup compared with classical TDA^{101,171}.

Other ways of obtaining higher-order latent relationships from single-cell multi-omics data require tensor decomposition methods¹⁷². Quantum tensor decomposition methods have been proposed that have provable quartic speedup with respect to classical counterparts¹⁷³, with potential implementation in PFTQD for biomedical data analysis¹⁷⁴ (Table 1) using variational quantum algorithms or by adapting a quantum approximate optimization algorithm¹⁷⁵. These bio-topological techniques applied to spatial single-cell omics data would make possible the capture of those cryptic interactions, which would have been hidden with even complete biological measurements, let alone with incomplete measurements encountered in practice, to better inform the spatiotemporal models described above.

Learning from limited samples

Despite the wealth of information generated from single-cell assays, including spatial and temporal, the availability of biological samples, particularly human samples, remains limited. Consequently, capturing cellular and molecular diversity across diseases and patient groups is challenging, hindering the development of models that reflect diverse phenotypes and populations. Even with acquired samples, the cost to perform single-cell or spatial assays is prohibitive for large-scale studies without large consortium efforts. The Human Cell Atlas^{7,8} contains data from approximately 63 million cells from just over 9,000 donors for a gene number of ~20,000 and an even larger number of proteins, making the problem hard to solve classically as there are far more variables than observations.

Assumptions about generalizing from limited biological samples remain common, but as high-resolution technologies expand the feature space, there is a growing need for methods that better address missing data and limited sample size. Imputation and generative tools are natural avenues to mitigate some of these challenges. There have been several recent efforts for imputation of data dropout in scRNA-seq¹⁷⁶. These tools are also being deployed in the generation of scRNA-seq data to overcome the limitations of low availability of biological samples¹⁷⁷. Quantum GANs have been used to generate classical data^{144,178}, aid in small molecule design^{146,179} and generate images¹⁸⁰, as well as to learn the underlying probability distributions to enable efficient loading into a quantum state¹⁴⁵. These methods enable learning data distributions with fewer parameters than classical GANs and have the potential of searching for an exponentially growing space that would be difficult for analysis by classical methods.

Effective training of classical ML algorithms requires large amounts of data, but when there are too few training samples as often occurs in biology, we can look towards to quantum computation as an alternative. In highly data-constrained settings, QNNs may offer certain advantages. Certain QNN architectures have been shown to have greater expressivity than some of their classical counterparts, allowing them to capture more complex probability distributions and indicating to potential speedups in training time¹⁸¹. Moreover, QCNNs, a subclass of QNNs, possess valuable characteristics for QML, including variational parameters scaling logarithmically in qubit count, resistance to barren plateaux during training⁶⁹ and potential to generate lower levels of generalization errors even in the case of limited training data⁵². In practice, we see that a QCNN can achieve comparable performance to CNNs while using orders of magnitude fewer trainable parameters^{70,182}, lending towards reduced risk of overfitting and greater generalization potential. However, a considerable hurdle to the practical application of certain QML algorithms is the nature of data requirements and the associated 'state preparation' overhead. Unlike classical ML, QML

requires classical information to be encoded into a quantum state, that is, state preparation, which can be computationally intensive and overshadow any gain from the quantum computation^{183–185}. Addressing this challenge requires innovative data encoding and state preparation techniques that minimize circuit complexity and will probably require domain knowledge to yield an effective heuristic for circuit design.

A quantum-enabled multimodal cellular model use case

As each of the computational threads that contribute to the development of a spatiotemporal cellular model (Fig. 2) is advanced through both classical and quantum computing techniques, we can look towards applications that would benefit from the combinations of techniques to gain deeper biological insights. One example would be applications for cell-based therapeutics¹⁸⁶. Although still considered the gold standard, the target-centric approach to drug discovery appears to be falling short on delivering substantial numbers of therapeutic advances¹⁸⁷, which may be attributed to the complexity of and uncertainty in target validation and off-target effects^{188,189}. A cell-based approach to understanding perturbation responses and disease behaviours presents a unique opportunity for multiple quantum and quantum-classical optimization and ML techniques to be brought together to address these challenges and further advance cell-centric therapeutic strategies. Figure 3 depicts how three research trajectories may capture complementary aspects of the cell-based approach, converging in a cancer use case. These include methodologies for engineering cells for immunotherapy, developing models of spatiotemporal cellular behaviour and extracting *d*th order feature interactions that would inform cellular behaviour. Each of these threads require advancements in the different computational efforts described in Fig. 3 and Table 1.

Designing cells for immunotherapy, such as chimeric antigen receptor (CAR)-T cells, is fundamentally a sample-constrained problem because of the high cost and time to generate these cells and the relatively large combinatorial space of intracellular signalling domains used in their construction. QCNNs could address this data limited problem and may scale better on quantum devices than QNNs. However, their applicability, as well as that of other quantum variational ML methods, might be constrained by challenges with state preparation, deep circuit structures more susceptible to accumulation of noise, and extended quantum device usage. These limitations could render such methods less practical for larger, more expressive circuits in PFQTD. In such cases, alternative hybrid classical-quantum methods may be necessary. For instance, projected quantum kernels^{190,191} combines quantum encoding of features with classical ML algorithms to learn with relatively shallow circuits and without the need of performing optimization of the circuit parameters.

Spatiotemporal cellular models require many of the computational tools previously described to capture and characterize the cells and cell types in a tumour microenvironment, from which models can be developed to predict cellular behaviour as the disease progresses or a therapeutic is introduced, such as a CAR-T cell. GNNs or autoencoders may be best suited for the ingestion of data and embedding into a latent space. With this latent representation a quantum algorithm can be incorporated to determine whether the performance can be enhanced. Quantum algorithms such as QCNN, VQC or quantum support vector machines can be used to train a variational model, albeit with certain challenges as noted above. Alternatively, methods such as quantum doubly stochastic transformers can be applied, which uses quantum computing for a specific portion of the transformer

Glossary

Ansatz

A parameterized quantum circuit used to approximate a target state or operation. In variational algorithms, its design and initialization crucially affect the accuracy and efficiency of the solution.

Continuous-time quantum walk

The quantum analogue of random walk, in which the evolution of a quantum system over a graph is continuous in time offering advantages in exploring complex networks and solving combinatorial problems.

Cumulants

Statistical measures that generalize moments to quantify higher-order correlations and dependencies among random variables. They isolate intrinsic multiway interactions, capturing nonlinear and non-Gaussian structure that ordinary moments or pairwise correlations cannot represent.

Curse of dimensionality

The exponential increase in data sparsity and computational complexity as the number of features grow. This phenomenon makes it harder for algorithms to detect meaningful patterns and they often require vastly more data for reliable analysis.

Dynamic circuits

A quantum circuit that incorporates classical processing within the coherence time of the qubits permitting mid-circuit measurements to enable feedforward operations such as using measured values to determine the next gates to be applied.

Ensemble learning

A machine learning technique that aggregates multiple individual models (known as ‘weak learners’) to produce a strong overall model with high prediction accuracy and robustness.

Entanglement

A uniquely quantum phenomenon, in which the states of two or more particles become interdependent; consequently measurement of one entangled particle instantaneously affects the state of the other particle.

Exponential speedup

A performance improvement of an algorithm whereby running time decreases exponentially compared to the state-of-the-art algorithmic benchmark.

Fidelities of 1-qubit and 2-qubit gates

Accuracies in quantum operations that create entanglement between qubits in a 2-qubit gate. If reduced, the effectiveness of quantum computations is decreased, and the complexity of solved problems is more limited.

Graph neural networks

(GNNs). A class of neural networks designed for graph-structured data, which aggregate and update node features based on neighbouring nodes while learning both local and global graph properties.

Kernel methods

A class of algorithms that use kernel functions, enabling them to transform data features to a high-dimensional, implicit feature space without exactly computing coordinates in the transformed feature space.

k-means

An unsupervised algorithm that partitions data into k clusters by assigning each point to the nearest centroid. k -means iteratively minimizes within-cluster variance and is widely used in unsupervised learning.

Laplacian

A matrix representation of graph connectivity that combines degree and adjacency matrices of the graph, enabling spectral clustering and dimensionality reduction of graphs.

Logical qubits

A higher-level abstraction of qubits used in a fault-tolerant quantum device where a single qubit is encoded using a collection of physical qubits while protecting quantum information from errors.

Neutral atom qubits

Qubits encoding quantum information in the internal states of individual, electrically neutral atoms held in optical or magnetic traps, offering long coherence times and scalability for quantum computing applications.

Nondeterministic polynomial-hard

A class of computational problems that are at least as hard as problems in the nondeterministic polynomial complexity class, meaning that finding an efficient solution for them would imply efficient solutions for all problem in nondeterministic polynomial.

Non-negative matrix factorization

A linear algebraic method to factor a data matrix V to two lower-rank non-negative matrices W and H representing parts of the original data matrix.

Optimal transport

A mathematical framework for finding the most cost-effective map from one probability distribution to another.

Quantum approximate optimization algorithm

A quantum algorithm that produces approximate solutions for combinatorial optimization problems.

Quantum circuits

Structured sequences of quantum gates operating on qubits to perform quantum computation.

Quantum convolutional neural networks

(QCNN). The quantum analogue of convolutional neural networks, in which quantum circuits perform convolution and pooling operations; it efficiently extracts hierarchical features from quantum data and has applications in quantum state classification and error correction.

Quantum network medicine

An emerging interdisciplinary field that leverages quantum computing and network science to analyse complex biological systems by applying quantum-enhanced techniques to biological interactions and reveal insights into disease mechanism and biomarker discovery.

Quantum neural network

(QNN). A quantum machine learning model that utilizes parameterized quantum circuits inspired by classical neural networks.

Qubits

The basic units of quantum information, analogous to classical bits. Unlike bits, qubits exist in a combination or superposition of 0 and 1 simultaneously, providing the foundation for quantum algorithms that can process information leveraging fundamentals of quantum mechanics.

Random walk

A stochastic process in which a walker on a graph starts at a vertex and repeatedly moves to one of its neighbouring vertices, chosen at random, generating a path through the vertices.

Sample limited

Refers to problems that are highly constrained in the number of samples, particularly with respect to the size of the feature space.

Glossary (continued)

Spin qubits

Qubits storing quantum information in the intrinsic spin states of particles like electrons or nuclei, often implemented in semiconductor quantum dots and controlled through magnetic or electric fields for quantum operations.

Superconducting qubits

Qubits that can exist in multiple states simultaneously based on superconducting circuits operating at extremely low temperatures to minimize thermal noise and maintain coherence, enabling fast gate operations for high-speed quantum computations.

Superposition

A fundamental principle in quantum mechanics stating that a quantum system can exist in multiple states simultaneously.

Tensor decomposition

A generalization of matrix factorization to multidimensional arrays or tensors, decomposed into latent lower-dimensional factors that capture key patterns in complex data.

Topological data analysis

(TDA). A mathematical method rooted in algebraic topology that analyses the inherent structure of data to reveal clusters, holes and voids in high-dimensional datasets that conventional data analysis techniques typically overlook.

Topological qubits

Quantum bits encoded in non-local properties of topological quantum systems, making them inherently resistant to local noise and decoherence.

Transformer networks

A deep learning architecture that uses self-attention mechanisms to capture long-range dependencies within a sequence and to draw inference.

Trapped-ion Qubits

Qubits that encode information in the electronic and nuclear spin states of individual ions, controlled using electromagnetic fields enabling long coherence times, making them suitable for high-accuracy quantum operations.

Variational quantum classifiers

(VQC). A quantum machine learning model that uses parameterized quantum circuits to classify data, with parameters optimized through a classical optimization method to minimize a cost function.

model and has shown benefit for vision transformers⁷¹. Such hybrid classical–quantum methods may offer a path to developing more accurate models.

Both cell engineering and spatiotemporal modelling would be enhanced by capturing the high-order interactions that would be present in the data given the complexity of biological systems and the hidden correlations present in biological data. These high-order latent interactions can be detected using quantum cumulants, QTDA or quantum tensor decomposition. Such methods capable of taking advantage of structures in the data would enable the capture of complex, hidden interactions that may be overlooked even if more complete data were available. The confluence of these methods can be used to develop cell-based precision medicine by modelling disease microenvironments at single-cell resolution to identify therapeutics, biologics or other interventions that modify the cellular disease ecosystem to make it more responsive to therapy or shift cells to a quiescent or even dying state.

Conclusion and future perspective

Although the state of quantum computing has progressed rapidly within the past ten years from small scale experiments with a few qubits to utility scale quantum computers with over 100 qubits commercially available, there remain key challenges that require considerable consideration ahead of near-term applications and careful planning for long term success given the trajectory of quantum technologies (Fig. 1). It is expected that in four years, with 200 error-corrected logical qubits capable of running 100 million gates will be available (as laid out in the IBM Quantum roadmap)^{192,193}. This advancement will be complemented by similar efforts to scale quantum devices spanning various technologies such as superconducting, trapped-ion, neutral atom and photonic qubits, by a host of companies and institutions¹⁹³.

These systems will substantially improve in error correction, coherence times, gate fidelities, execution times and qubit quality towards developing fault-tolerant quantum systems^{194,195} and quantum centric supercomputers^{194,196,197}, which would seamlessly integrate AI, high-performance computing and quantum computers. Although there is clear promise in the future of quantum computing hardware, we must account for its current limitations and remain adaptable to this rapidly evolving technological field where new insights and challenges are being discovered routinely.

State-of-the-art variational QML algorithms can be challenged to learn at scale, despite advancements in error mitigation, fractional gates and dynamic circuits. Within QML, the relatively limited number of qubits or circuit execution times often necessitates operating in a lower-dimensional feature or sample space, respectively, through methods such as subsampling or projection, which may introduce their own biases or information loss. Methods such as quantum support vector machines can have their circuit depth grow dramatically with qubit numbers, which can quickly exceed the coherence times of available quantum systems. In some cases, the learning task can be reformulated into an optimization task using methods such as quantum approximate optimization algorithms^{198–200} or using a hybrid quantum–classical task such as projected quantum kernels^{190,191}, both of which have found success on PFTQD owing to features such as relatively shallow circuits. Additional hybrid quantum–classical methodologies, which may use a classical AI-based method to generate lower-dimensional embeddings and variational quantum algorithms to generate quantum feature maps on which to perform prediction and classification tasks, have shown utility in PFTQD^{201,202}.

Despite their popularity in PFTQD, variational algorithms have limitations such as barren plateaux, where gradients vanish exponentially during the optimization process with growing system size²⁰³.

Another challenge is the choice of *ansatz* in these algorithms, where a hardware-efficient *ansatz* may not accurately represent the quantum algorithm modelled on the data or may be impractical in PFTQD²⁰⁴. As quantum hardware matures, quantum algorithms operating in native data spaces may outperform classical methods, enabling solutions to problems with sharp complexity phase transitions, such as high-order computations in tensor decompositions, TDA or cumulants, which are beyond the grasp of classical computing¹⁸⁶.

Alongside quantum computing, there are ongoing efforts in quantum-inspired and dequantized algorithms to identify quantum advantages and adapt them to enhance classical methods without the overhead of quantum state preparation^{205,206}, thereby removing theoretical exponential speedups. The bulk of dequantized algorithms pertain to quantum linear algebra algorithms that assume a low-rank input data given classically²⁰⁷. This low-rank assumption enables classical algorithms to achieve similar performance to quantum algorithms in the form of randomized algorithms that permit matrix sketching and dimensionality reduction akin to the quantum state preparation of quantum algorithms^{205,206}. These dequantization methodologies can still permit large polynomial speedups, which in practice could have considerable benefits for the healthcare and life sciences communities. The quantum-inspired algorithms related to the dequantization efforts have indeed shown an exponential speedup, though at the cost of a large polynomial overhead and of quickly degrading performance as the dimension, rank and condition number of the input matrix increases²⁰⁵.

Furthermore, any performance assessment of a quantum algorithm will require rigorous testing and benchmarking. Such benchmarking can focus on classes of problems that are known to be difficult to solve classically and therefore represent opportunities for a quantum algorithm to perform better, as has been proposed for optimization problems²⁰⁸. A similar benchmarking suite would prove invaluable for other classes of quantum algorithms, including QML and quantum simulation, to demonstrate where quantum computing may have the most impact. It is evident that the field of quantum computing is rapidly evolving to understand under what conditions it can offer meaningful performance gains for healthcare and life science, gains that may be measured in faster computation, whether exponential or polynomial, or in increased accuracy.

Single-cell and spatial single-cell technologies have opened a vast array of opportunities to advance biological insights and therapeutic design. These opportunities have been made possible by tremendous advancements in computational tools and techniques. These tools have been entirely classical up to this point, but as quantum computing is rapidly developing, we are now at a point where exploring how quantum computation can work together with classical computation to promote new discoveries is a reality. Quantum computing has the potential to become a powerful enabling technology to help push single-cell, spatial single-cell and new therapeutics models forward.

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Author contributions

The authors contributed equally to all aspects of the article.

Competing interests

The authors declare no competing interests.

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